'compressed' Linux kernel. With the development of virtual memory, the prefix *vm* was used to indicate that the kernel supports virtual memory.

- *Initrd.img-3.2.0-29-generic:* An initial 'ramdisk' for your kernel.
- *Config-3.2.0-29-generic:* The 'config' file is used to configure the kernel. We can configure, define options and determine which modules to load into the kernel image while compiling.
- *System.map-3.2.0-29-generic:* This is used for memory management before the kernel loads.

Kernel modules

The interesting thing about kernel modules is that they can be loaded or unloaded at runtime. These modules typically add functionality to the kernel—file systems, devices and system calls. They are located under */lib/modules* with the extension *.ko*.

Setting up a development environment

Let's set up the host machine with Ubuntu 14.04. Building the Linux kernel requires a few tools like GIT, *make*, gcc and *ctag/ncurser-dev*. Run the following command:

```
Sudo apt-get install git-core gcc make libncurses5-dev
exuberant-ctags
```

Once GIT is installed on the local machine (I am using Ubuntu), open a command prompt and issue the following commands to create an account:

```
git config --global user.name "Vinay Patkar"
git config -global user. Email Vinay_Patkar@dell.com
```

Let's set up our own local repository for the Linux kernel.

> **Note:** 1. Multiple Linux kernel repositories exist online. Here, we pull Linus Torvald's Linux-2.6 GIT code -- Git clone *http://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux-2.6.git*
> 2. In case you are behind a proxy-server, set the proxy by running *git config –global https.proxy https://domain\usernmae:password@proxy:port.*

Now you can see a directory named *linux-2.6* in the current directory. Do a *GIT pull* to update your repository:

```
Cd linux-2.6
Git pull
```

> **Note:** Alternatively, you can clone the latest stable build as shown below:
>
> ```
> git clone git://git.kernel.org/pub/scm/linux/kernel/git/
> stable/linux-stable.git
> cd linux-2.6
> ```

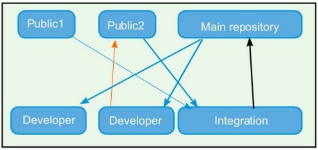

Figure 1: GIT

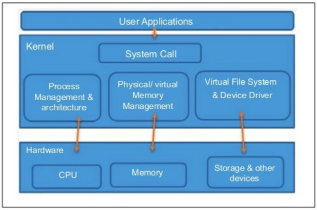

Figure 2: Linux kernel architecture

Figure 3: Locating Ubuntu Linux kernel files

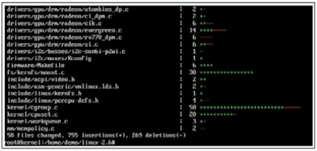

Figure 4: GIT pull

Next, find the latest stable kernel tag by running the following code:

```
git tag -l | less
git checkout -b stable v3.9
```

> **Note:** RC is the release candidate, and it is a functional but not stable build.

Once you have the latest kernel code pulled, create your own local branch using GIT. Make some changes to the code and to commit changes to the code, run *git commit –a.*